

HAO HU

AI & Agentic Systems Engineer · Staff-Level Engineering Leader

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SUMMARY

Staff-level software engineer with 10+ years at Google, Uber, and Microsoft building large-scale distributed systems, data infrastructure, and platforms — now applying that foundation to production AI/agentic systems as a startup CTO.

Builds production AI systems — RAG, AI data analytics, and tool-using agents with deterministic guardrails, citation-grounded refusal, human approval for risky actions, and golden-set regression evals.

Hands-on engineering leader with strong CS fundamentals and architecture judgment — leads small teams from architecture through delivery, using spec-driven AI-assisted workflows to ship faster without sacrificing reliability.

TECHNICAL SKILLS

LLM & Agents: agent orchestration, tool / function calling, MCP integration, workflow/state graphs, multi-agent routing, context engineering / memory, LangChain / LangGraph / DeepAgents, OpenAI / Claude / DeepSeek

AI-Native Engineering: spec-driven development, AI-assisted implementation, TDD / validation, review-gated output, harness engineering, Claude Code / Codex / Cursor / OpenCode

RAG & Retrieval: RAG (multimodal), hybrid retrieval, embeddings, reranking, query optimization, Milvus, Qdrant

Evaluation & Safety: eval harnesses, golden-set regression, RAGAS, LLM-as-judge, guardrails, human-in-the-loop

Inference & ML: PyTorch, vLLM, SGLang, Ollama, Hugging Face, XGBoost / LightGBM / CatBoost

Languages: Java, Go, Python, SQL, C++, C#

Backend & Distributed Systems: microservices, API design, gRPC / Protobuf, REST, Thrift, Spring AI, FastAPI, SSE

Data Systems: data modeling, CDC, PostgreSQL, MySQL, Redis, BigQuery, Kafka, Flink, ClickHouse, MongoDB

Cloud & Platform: Google Cloud (Cloud SQL, GCE, GCS), Docker, Kubernetes, CI/CD, observability

EXPERIENCE

Yuanbao Creative Technology

Feb 2025 – Present

Chief Technology Officer

Wuhan, China

- **Architected and shipped a unified ERP AI layer** spanning operations, analytics, and knowledge, built on an evaluation-first practice: golden sets, regression suites, least-privilege tools, and human approval for risky actions.
- **Operations Copilot:** built a **DeepAgents**-based agent harness over ERP entities (product / inventory / order / supplier / PO), using governed **MCP** tools and a Spring Boot / Spring AI server to enforce approval-gated writes.
- **Analytics Agent:** delivered self-service NL-to-SQL over the ERP **ClickHouse** warehouse, backed by a **Qdrant** semantic layer, deterministic **SQLGlot** guardrails, scoped read-only execution, and bounded SQL repair.
- **Knowledge Assistant:** delivered **LangGraph**-orchestrated RAG over enterprise documents with **Milvus** hybrid retrieval and intent-specific pipelines; lifted answer quality with query expansion, HyDE, RRF/rerank, a retrieval test bench, and golden-set feedback loops, while enforcing citation grounding and strict-evidence refusal.

Uber

May 2022 – Mar 2023

Senior Software Engineer

Toronto, ON

- **Built Uber's first CPG (consumer-packaged-goods) Ads feature — storefront banner ads** — as backend lead on a 3-person team, delivering a backbone microservice bridging the UberEats storefront and the Uber Ads platform so advertisers run budgeted creative for chosen items and brands across markets; **scaled to ~250K QPS at 180 ms p99** within a quarter and onboarded marquee advertisers (Coca-Cola, Red Bull). (Go, gRPC / Thrift, DocStore.)
- **Redesigned the ad-campaign data model for CPG scale:** replaced the restaurant-era single-location, item-ID-keyed model — which needed thousands of entries and a 4-person service team to run one national campaign — with a GTIN-keyed model (Global Trade Item Number) bundling brand, city, and country, eliminating millions of duplicate campaign entries and the manual scaling burden for sales.
- **Launched store-menu sponsored items** via a third-party ad integration: as Ads-team dev representative, sourced sponsor candidates, prioritized sponsored items in storefront sub-menus, and streamed view/click/purchase signals back for billing through a Kafka + Flink pipeline.
- **Served as central dev representative** across data scientists, sales, DevOps, partner teams, and the third-party provider, and built monitoring and alerting (Prometheus / Grafana) into both launches.

Google

Feb 2018 – Feb 2021

Software Engineer — Cloud SQL (Data Integration / Disaster Recovery)

Sunnyvale, CA

- **Owned and launched Cloud SQL data export / import:** delivered a Postgres CSV export/import feature to GA with permission controls, resolved long-standing delimiter/format corruption via [customizable delimiters](#), and secured the Google Cloud Storage–interacting flows with Cloud SQL IAM.
- **Built the backup verification pipeline:** validated daily Persistent Disk snapshots against Cloud SQL / Spanner metadata by attaching them to pooled instances, booting MySQL / PostgreSQL, and checksum-comparing, while pruning snapshots past retention.
- **Raised backup health from 99.992% to 99.997%** across a large fleet backing a 99.95% SLA, by root-causing and fixing the top causes of backup corruption (chiefly disk exhaustion from unrotated general logs).
- **Enabled Point-in-Time Recovery (PITR):** built general-log cleanup and introduced transaction-log retention so any MySQL instance could be restored to an arbitrary timestamp (nearest backup + transaction-log replay).
- **Redesigned the backup retention policy** to always keep the most recent *healthy* backup (vs. a flat 7-backup window), guaranteeing every instance had a recoverable backup.
- **Mentored junior engineers** and served on the on-call rotation, resolving release-blocking and customer-escalated issues. (Java backend, Go on-instance.)

Microsoft

Jan 2014 – Oct 2017

Software Development Engineer

Mountain View, CA

- **Independently designed and built a distributed tool to profile 400M+ legacy Hotmail accounts** (C++), collecting account characteristics so the Outlook migration's account selector could stage cutovers and close 40+ feature gaps; partnered with the migration team through a successful migration.
- **Co-built “People I Communicate With” (PICW) from the ground up** on a 3-person team — an Outlook component (C#) that efficiently captures and stores user communication signals (frequent contacts; mail read-and-ignored, left unread, junked, or reported as spam/phishing), initially powering Contact Safety and anti-spam.
- **Took sole ownership of PICW and scaled it**, feeding signals into the People Relevance people-graph and serving as dev representative across teams; enabled by default in Outlook on the web, IMAP, and POP for enterprise and multi-tenant customers (100M+ Office 365 commercial users in 2017).
- **Served on the on-call rotation**, resolving release-blocking and customer-escalated issues.

Epic Systems

Jun 2012 – Nov 2013

Software Developer

Madison, WI

- **Built and maintained EMC2**, Epic's internal development-management platform — artifact tracking, source control, and deployment monitoring (an early CI/CD system) — in VB.NET on InterSystems Caché.
- **Developed customer release and upgrade utilities** (console and web) for software upgrades and version checks, streamlining how customers kept their deployments current.
- **Provided developer enablement and escalation support**, unblocking development and deployment issues and delivering a Trial Dev Environment service built on a multi-way merge for safe change validation.

HCR ManorCare

Jan 2011 – Apr 2012

Data Warehouse Developer

Toledo, OH

- **Built SSIS ETL pipelines** loading multi-source data into an enterprise star-schema data warehouse, with validation, logging, and error handling.
- **Developed advanced SSRS reports** (cross-tab, drill-down, parametric, cascaded) and T-SQL stored procedures delivering finance and operations business intelligence.

EDUCATION

Bowling Green State University

M.S. in Computer Science

Aug 2010 – May 2012

GPA 4.0/4.0 · Bowling Green, OH

Wuhan University

B.S. in Software Engineering

Aug 2006 – Jun 2010

GPA 3.5/4.0 · Wuhan, China